From Housing Gaps to Full Living-Cost Modeling

Updated affordability analysis for EdTrust

Introducing food and transportation assumptions, institutional ranking, and predictive modeling for a comprehensive affordability framework.

Project Evolution: Beyond Housing-Only Gaps

Expanded Scope

Moved from descriptive housing analysis to a comprehensive affordability modeling framework.

Integrated Assumptions

Incorporated evidence-based food and transportation costs to complete the living-cost picture.

Advanced Analytics

Shifted focus to institutional ranking, regression modeling, and equity-focused Pell analysis.

Key Goal

Provide a more realistic estimate of the total affordability gap faced by students.

The Updated Affordability Framework

STEP 1: CALCULATE TOTAL LIVING COST

HUD Housing + Food Assumption + Transportation Assumption = Modeled Total Living Cost

STEP 2: DETERMINE AFFORDABILITY GAP

Modeled Total Living Cost - Reported Living Cost = Modeled Total Gap

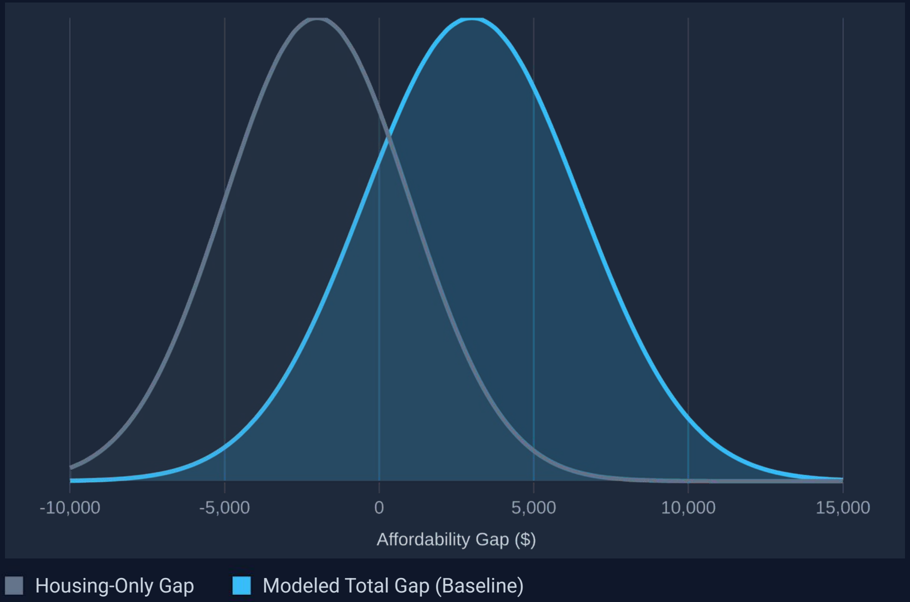
- **Reported Living Cost:** Derived from IPEDS Cost of Attendance (COA) minus tuition and fees.
- **Alignment Strategy:** Grounding institutional reports in local cost benchmarks for transparency.

Modeling Assumptions & Sensitivity Scenarios

SCENARIO	FOOD (MONTHLY)	TRANSPORT (MONTHLY)	9-MONTH TOTAL	PRIMARY USE CASE
Low	$291	$100	$3,519	Walkable / transit-heavy settings
Baseline MAIN SPEC	$360	$200	$5,040	National student default
High	$449	$350	$7,191	Car-dependent / high-mobility settings

** Note: These are modeled assumptions based on regional benchmarks, not institution-reported values.*

Distribution of Modeled Total Gaps (Baseline)



Upward Distribution Shift

Incorporating food and transportation costs shifts the entire distribution significantly upward compared to housing-only models.

Alignment Reassessment

Institutions previously seen as overestimating may now appear closer to alignment or underestimation under the full living-cost lens.

Estimated Mismatch

The baseline scenario reveals a broader "estimated living-cost mismatch" across the higher education sector.

Sensitivity Analysis: Interpreting the Range

SCENARIO	COMPONENTS INCLUDED	ESTIMATED GAP IMPACT
Housing-Only	HUD Rent only	Baseline reference
Low Scenario	Rent + $3,519 (Food/Transport)	Moderate upward shift
Baseline Scenario	Rent + $5,040 (Food/Transport)	Primary specification
High Scenario	Rent + $7,191 (Food/Transport)	Significant upward shift

Key Takeaway: Results should be interpreted as a **range of estimates** rather than a single exact figure. The direction and size of affordability gaps are sensitive to regional and behavioral cost assumptions.

Institutional Ranking: Identifying Patterns

Bright Spots

Institutions where modeled total costs most closely align with reported living costs. These schools show a gap near zero, indicating high transparency in cost reporting.

Positive Gaps

Cases where reported costs are significantly lower than modeled local living benchmarks. This suggests potential underestimation of non-tuition expenses.

Negative Gaps

Cases where reported costs exceed modeled total living costs. These institutions may be overestimating local living requirements in their financial aid calculations.

Public Utility: Ranking serves as the primary policy-facing output for identifying sector-wide alignment trends and informing student choice.

Regression Model: Predicting Affordability Gaps

MODEL SPECIFICATION

Dependent Variable (Outcome)

Modeled Total Affordability Gap

INSTITUTIONAL PREDICTORS

- Pell Grant Recipient Share
- Urbanicity Group
- Carnegie Classification
- BEA Economic Region

KEY ANALYTICAL INSIGHTS

Purpose: Estimate which institutional characteristics are most strongly associated with larger or smaller modeled total affordability gaps.

Methodology: Ordinary Least Squares (OLS) regression using the baseline scenario specification.

Exclusion Rule: Local rent is excluded as a predictor because it is already a component of the dependent variable, preventing circularity.

Results identify statistical associations, not causal relationships or institutional motives.

Logistic Classification: Likelihood of Underestimation

BINARY OUTCOME VARIABLE

Is the Modeled Total Gap Positive?

MODEL PREDICTORS

- Pell Share
- Urban Group
- Carnegie Group
- BEA Region

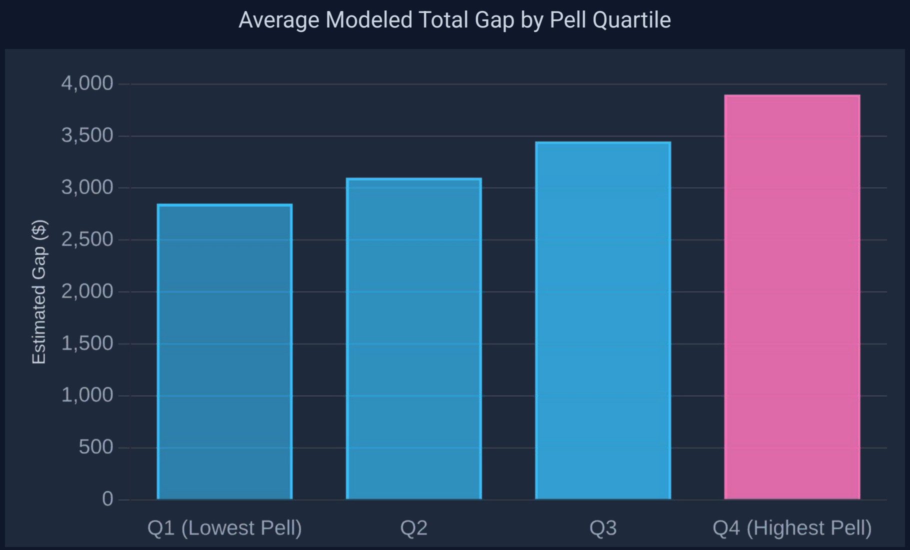
INTERPRETATION

Positive Gap Meaning: Modeled total living costs exceed the institution's reported living costs.

Analytical Goal: Identify systematic patterns where institutions are more likely to have estimated living-cost mismatches.

This model identifies statistical patterns of higher likelihood, not institutional intent or proof of misconduct.

Equity Lens: Analysis by Pell Share



Pell Quartile Analysis

Shows how modeled total gaps vary across institutions by share of low-income students.

Equity Connection

Checks if high-Pell colleges show different cost-alignment patterns.

Policy Focus

Frames an exploratory equity lens to spot patterns needing targeted policy action.

Recommendations for Final Deliverables

Institutional Ranking Table

A policy-facing tool allowing stakeholders to filter by region, urbanicity, and Pell share to identify alignment trends.

Policy Brief

A concise summary of baseline results, sensitivity ranges, and key equity insights for decision-makers.

Interactive Dashboard

An optional visual platform for exploring the "modeled total affordability gap" across the U.S. higher education landscape.

CORE PRINCIPLES

- ✓ Maintain transparency regarding assumptions
- ✓ Emphasize alignment over institutional motive
- ✓ Ground metrics in student lived reality

Limitations and Careful Interpretation

MODELED VS. OBSERVED

Food and transportation are evidence-based **modeled assumptions**, not observed institution-specific data points.

DATA CONSTRAINTS

COA data is missing for many institutions, resulting in a smaller analytic sample than the full merged dataset.

ALIGNMENT, NOT INTENT

Results describe **alignment patterns** with local benchmarks, not institutional intent or "misleading" behavior.

ESSENTIAL CONTEXT

Sensitivity analysis is mandatory; assumptions significantly affect the size and direction of the estimated gaps.

Focus on "estimated living-cost mismatch" and "policy-facing ranking" rather than causal claims or proof of misconduct.